

CELL AND MOLECULAR BIOLOGY LABORATORY

From Gene Mutation to Disease

Investigating How DNA Sequence Changes Affect Protein Products and Human Phenotypes

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Part I. Disease Background

Hemophilia B is an inherited bleeding disorder characterized by a deficiency or abnormal function of coagulation Factor IX (FIX), a protein that plays an important role in the blood-clotting process. It is also known as Christmas disease and is considered the second most common type of hemophilia, accounting for approximately 15–20% of hemophilia cases. Hemophilia B primarily affects males because the gene responsible for the production of Factor IX is located on the X chromosome.

The genetic cause of Hemophilia B is associated with mutations in the F9 gene, which is located on the long arm of the X chromosome at Xq27.1. The F9 gene provides the instructions for producing coagulation Factor IX. Mutations in this gene can result in reduced production or impaired function of Factor IX, leading to an inability of the blood to form a stable clot efficiently. Different types of pathogenic variants, including missense, nonsense, splice-site, insertion, deletion, and promoter variants, have been identified in the F9 gene. The type and effect of the mutation can influence the amount of functional Factor IX available in the blood and, consequently, the severity of the disorder.

Part II. Research the Disease

A. Disease

13. What is the disease or phenotype?	Hemophilia B (F9), once known as Christmas disease, a rare inherited blood disorder that impairs the body's ability to make blood clots which is essential for stopping bleeding. Caused by missing or faulty Factor IX (FIX).
14. What are its major clinical characteristics?	It causes prolonged bleeding, easy bruising, and bleeding into the joints and muscles.
15. Which cells, tissues, or organs are mainly affected?	The blood-clotting system is mainly affected, especially the joints and muscles.
16. What is its genetic basis?	It is caused by mutations in the F9 gene, which affects the production or function of factor IX.
17. What is its inheritance pattern, if applicable?	It is X-linked recessive, so it mainly affects males while females are often carriers.

B. Gene and Normal Protein

18. What is the official gene symbol?	The official gene symbol is F9
19. On which human chromosome is the gene located?	It is located in X chromosome (Xq27.1)
20. What does the gene normally encode?	The F9 gene encodes coagulation factor IX (FIX).
21. What is the normal biological function of the protein?	Factor IX is a clotting protein/enzyme that helps form a stable blood clot. When activated to FIXa, it helps activate Factor X during coagulation.

22. Where in the cell is the protein normally found?	The Factor IX is mainly found in the blood/plasma after being produced and secreted by liver cells.
23. In what biological pathway or cellular process does it participate?	It participates in the blood coagulation (clotting) pathway, particularly the intrinsic pathway.

C. Documented Mutation

Gene	F9
Reference Transcript	NM_000133.4
Exact Variant Notation	NM_000133.4(F9):c.676C>T
Nucleotide Change	C → T at nucleotide 676
Predicted Protein Change	NP_000124.1:p.Arg226Trp
Mutation Type	Missense mutation
ClinVar accession	VCV000010590.21
Clinical Interpretation	Pathogenic/Likely pathogenic
Scientific Reference	Suehiro, K., et al. (1989). Blood clotting factor IX BM Nagoya. Substitution of arginine 180 by tryptophan and its activation by α -chymotrypsin and rat mast cell chymase. Journal of Biological Chemistry, 264(35), 21257–21265. PMID: 2592373.

Hypothesis

If a mutation is introduced into the normal F9 coding sequence, it is expected to cause a change in the resulting factor IX protein depending on the type of mutation. The documented F9 c.676C>T mutation is expected to change one amino acid from arginine to tryptophan without shifting the reading frame, while the artificial one-nucleotide deletion is expected to cause a frameshift and alter the amino acid sequence after the deletion. These changes may affect the normal function of factor IX and help explain its relationship to Hemophilia B.

Methods

This study used Galaxy to examine the effects of mutations in the F9 gene by comparing the wild-type sequence with documented and artificial mutant sequences. The wild-type F9 coding sequence from NM_000133.4 was obtained and uploaded into Galaxy as the reference sequence. The sequence was translated using the Transeq tool to obtain the predicted factor IX protein. The WT sequence and its translated protein were examined for their sequence length, reading frame, and amino-acid sequence.

The documented F9 c.676C>T mutation was then reproduced by changing the C nucleotide to T at position c.676 in a copy of the WT coding sequence. The edited sequence was translated using Transeq and compared with the WT protein to identify the resulting amino-acid change and determine whether the reading frame or protein length was affected. The documented mutant sequence was also examined for the presence of a premature stop codon.

An additional artificial mutation was created by deleting one nucleotide at c.71 from the WT F9 coding sequence. The resulting sequence was translated and compared with the WT and documented mutant proteins. The three sequences were compared to observe differences in mutation type, amino-acid sequence, reading frame, protein length, and premature stop codons. The results were then used to describe how different types of mutations can produce different effects on the F9 protein and its possible function.

Part III. Obtain the Normal Reference Sequence

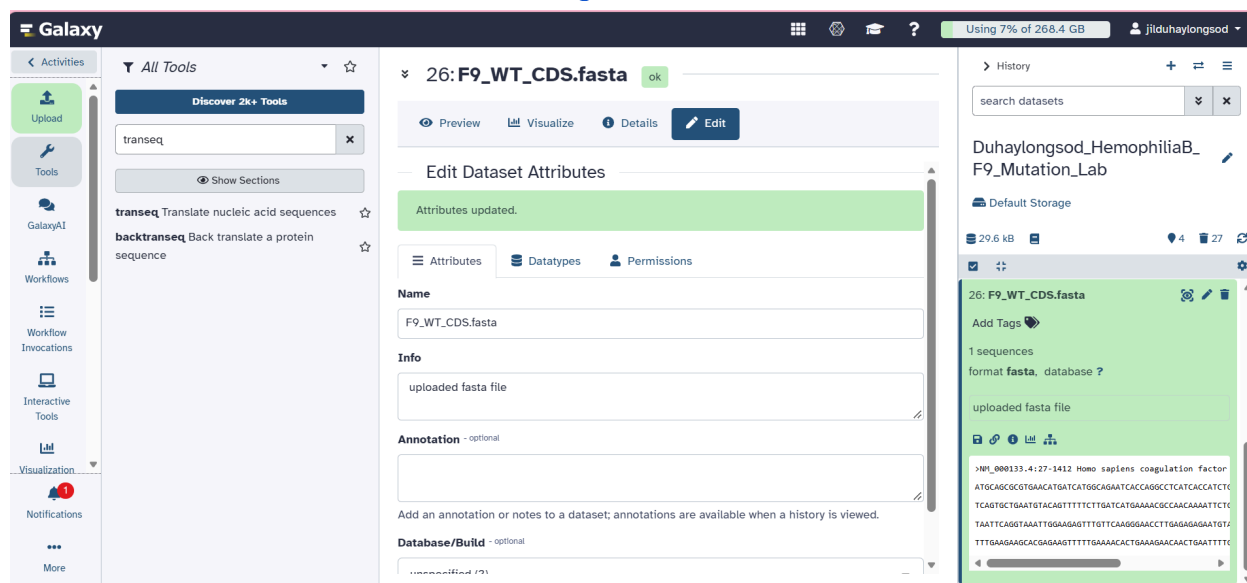


Figure 1. Renaming the wild-type F9 coding sequence (CDS) in Galaxy as F9_WT_CDS.fasta before translating it into the protein sequence.

Part IV. Import the Normal Sequence into Galaxy

The wild-type F9 coding sequence (CDS) was imported into Galaxy and designated as the reference for the analysis. A separate Galaxy history was established to keep the sequences, datasets, and analysis results organized. The original WT sequence was preserved without any modifications and was used as the baseline for comparison with both the documented and artificial mutations.

Part V. Establish the Wild-Type Control

CDS length	1,386 bp
Predicted protein length	461 aa
Start codon	ATG
Stop codon	TAA
Reading frame	Frame 1 (1+)
First 10 amino acids	MQRVNMIMAE
Last 10 amino acids	NWIKKTKLT

The screenshot displays the Galaxy web interface. On the left, the 'Tools' panel shows the 'transeq' tool selected under 'Discover 2k+ Tools'. The main panel shows the '27: F9_WT_protein.fasta' dataset with a 'Preview' button. The preview shows the mRNA sequence: >27-1412_1 Homo sapiens coagulation factor IX (F9), transcript variant 1, mRNA. The sequence is: MQRVNMIMAESPLTITICLLGYLLSAECTVFLDHNANKTLNRPKRYNSGKLEEFVQGNL ERECHKEKCSFEAREVEFENTERITTEFAKQYVGGQCESNPCLNGSSCKDDINSYECWP FGFEKNCLELDTVCHIKNGRCQFCNKSADNKKVCSCTEGYRLAENQKSCPEAPFPFCGR VSVSQTSLTRAETVFPDQVYNSTEAEITLONITQSTQSFNDFTRVGGEDAKPQQFPW QVVLNGKVDACGGSIVNIEKIVTAACHVETGVKITVAGEHNIEETEHTEQKRWIRII PHHRYNAAINKYNIHDIALLEDEPLVLSVYTPICIAQKEYTNIFLKFGSGYVSGMRVF HKGRSALVLQYLRVPLVDRATCLRSKFTIYNIMFCAGFHEGGDRSCQGDSSGPHVTEVE GTSFLTGIIISGEECAWKGKGYITVRSRYNMIKTKLT*. On the right, the 'History' panel shows a search for 'Duhaylongsod_HemophiliaB_F9_Mutation_Lab' and a list of datasets including '27: F9_WT_protein.fasta' and '26: F9_WT_CDS.fasta'.

Figure 2. Translation of the wild-type F9 coding sequence into the predicted coagulation factor IX protein using the Transeq tool in Galaxy.

Part VI. Formulate a Mutation Hypothesis

Mutation and exact nucleotide change	F9 c.676C>T
Number of nucleotide(s) affected	1 nucleotide
Predicted mutation type	Missense mutation
Predicted effect on the reading frame	No change in reading frame
Predicted effect on protein length	No change; expected to remain 461 aa
Predicted effect on protein function	The mutation is expected to change arginine to tryptophan at amino-acid position 226, which may alter the structure or function of factor IX.

Part VII. Create the Mutant Sequence

Original Nucleotide Position	c. 676
Original Sequence	C
Mutant Sequence	T
Number of bases inserted/deleted/substituted	1
Mutation type	Missense

Part VIII. Translate the Mutant Sequence

Mutant CDS Length	1,386 bp
Mutant protein length	461 aa
Reading frame	Frame 1 (1+)
Location of the first amino acid difference	Position 226
Presence or absence of a premature stop codon	Absent
Approximate number of amino acids affected.	1 amino acid
Mutant protein FASTA file name	F9_MT_CDS_protein.fasta

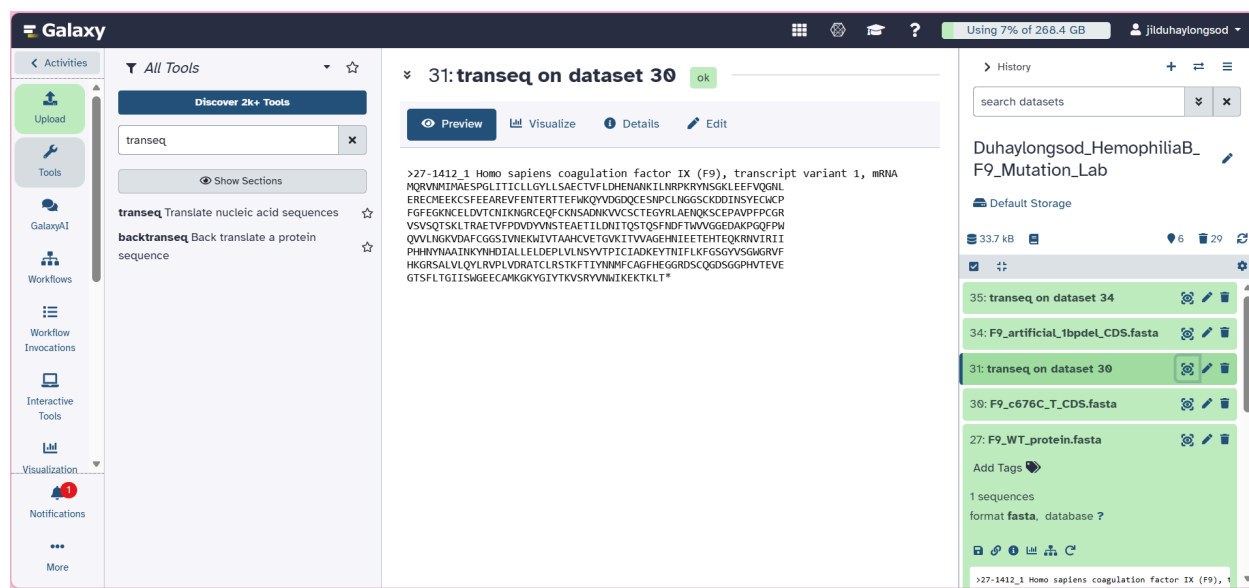


Figure 3. Figure X. Translation of the mutant F9 coding sequence using the Transeq tool in Galaxy, showing the predicted factor IX protein sequence after introducing the documented c.676C>T (p.Arg226Trp) mutation.

Part IX. Compare WT and Mutant Proteins

At what amino-acid position do the sequences first differ?	Position 226
Is only one amino acid affected?	Yes
Are multiple downstream amino acids changed?	No
Was an amino acid deleted or inserted?	No
Was a premature stop codon produced?	No
Did the reading frame change?	No
Did the protein length change?	No, both are 461 aa
Is the mutation missense, nonsense, frameshift, in-frame deletion/insertion, repeat expansion, or another type?	Missense mutation

The mutation changes the C base to T at nucleotide c.676, changing CGG to TGG and arginine to tryptophan at amino acid 226. The reading frame and protein length remain unchanged, confirming a missense mutation.

Part X. Explain the Molecular Consequence

F9 c.676C>T → C changes to T → CGG changes to TGG → R226 changes to W → factor IX structure/function may be affected → normal coagulation is impaired → blood clotting is reduced → prolonged/excessive bleeding → Hemophilia B.

The F9 gene contains the instructions for making coagulation factor IX (FIX), a protein that is important for normal blood clotting. Normally, the F9 DNA sequence is transcribed into mRNA, and the ribosome reads the mRNA three nucleotides at a time to produce the correct amino acid sequence. This results in a complete 461-amino-acid factor IX protein that can function in the blood-clotting process.

In the F9 c.676C>T mutation, one nucleotide is changed from C to T at position 676 of the coding sequence. This changes the codon from CGG to TGG. Because only one nucleotide is substituted, the reading frame is not shifted. However, the codon change causes arginine (R) at amino acid position 226 to be replaced by tryptophan (W), producing the p.Arg226Trp (R226W) protein.

The change from arginine to tryptophan may affect the structure or function of factor IX, which can reduce its normal activity in the coagulation pathway. As a result, the blood-clotting process may not work properly, leading to reduced clotting ability. This can cause prolonged or excessive bleeding, which is the characteristic phenotype associated with Hemophilia B.

Part XI. Second Experiment: Create Your Own Mutation

Artificial Mutation	One-nucleotide deletion
Nucleotide Position	c.71
Original Sequence	TCAGTG
Mutant Sequence	CAGTG
Number of bases affected	1 nucleotide base deleted
Predicted Mutation Type	Frameshift mutation
Predicted Effect in Reading Frame	The deletion of one nucleotide (c.71) shifts the reading frame from the deletion site downward, changing how the following nucleotides are grouped into codons.
Predicted Effect on Protein	The amino acid sequence will change from the deletion site onward and may produce a premature stop codon.

For the second experiment, I created an artificial mutation in the F9 coding sequence by deleting one thymine (T) at nucleotide position c.71 (c.71delT). Since only one nucleotide is removed, the reading frame is expected to shift from the deletion site onward, changing all the downstream codons. I predict that the translated factor IX protein will have a different amino acid sequence after the mutation and may terminate early because of a premature stop codon, producing a shorter and likely nonfunctional factor IX protein.

Part XII. Compare the Documented and Artificial Mutations

	WT Sequence	Documented mutation	Artificial mutation
CDS Length	1, 386 bp	1, 386 bp	1, 385 bp
Protein Length	461 aa	461 aa	Shorter/abnormal due to premature stop
Mutation Type	None	Missense	Frameshift
Reading frame changed	No	No	Yes
Premature stop codon appeared	No	No	Yes
Amino acids affected	None	1 amino acid (position 226)	Many amino acids from around position 24 onward.
Expected functional consequence	Normal factor IX protein	May alter factor IX function	Likely strongly affects factor IX because the frameshift changes the downstream amino-acid sequence and causes an early stop

The WT F9 sequence has 1,386 bp and produces a normal 461-aa factor IX protein. The documented mutation is a missense mutation that changes only one amino acid at position 226 without changing the reading frame or protein length. In contrast, the artificial mutation deletes one nucleotide, causing a frameshift, changing many downstream amino acids and producing a premature stop codon. Therefore, the artificial mutation is expected to have a stronger effect on the factor IX protein.

Part XIII. Interpretation Questions

32. Why does the exact location of a mutation matter?

- The exact location of a mutation matters because it determines which part of the gene or protein is affected. A mutation in an important coding region can change an amino acid and possibly affect protein function. In our experiment, the **F9 c.676C>T** mutation occurs in the coding sequence and changes amino acid 226 from arginine to tryptophan. ClinVar also reports that this amino acid residue is highly conserved and that the variant has been observed in people with hemophilia B.

33. Why can deleting three nucleotides produce a different result from deleting one or two nucleotides?

- Deleting three nucleotides can remove one complete codon without changing the reading frame because codons are made of three nucleotides. However, deleting one or two nucleotides shifts the reading frame, causing the downstream nucleotides to be grouped into different codons. This can change many amino acids and may produce a premature stop codon.

34. Does every mutation change the amino-acid sequence? Explain.

- No, not every mutation changes the amino-acid sequence. Some mutations are called synonymous mutations, where the changed nucleotide still produces the same amino acid because more than one codon can code for the same amino acid. Mutations outside the protein-coding region can also affect gene regulation without directly changing the protein sequence.

35. Does every amino-acid substitution destroy protein function? Explain.

- No, an amino-acid substitution does not always destroy protein function. Its effect depends on the location of the amino acid and the properties of the original and new amino acids. Some substitutions have little effect, while others can change protein structure or function. In our F9 mutation, arginine is replaced by tryptophan, and ClinVar reports this variant as pathogenic and associated with hemophilia B.

36. Why can a frameshift affect many amino acids even if only one nucleotide was deleted?

- A frameshift can affect many amino acids because deleting one nucleotide changes the grouping of the nucleotides into codons. After the deletion, the ribosome reads the downstream sequence in a different way, so many of the following codons can produce different amino acids. This can also result in a premature stop codon.

37. Why might a premature stop codon produce a nonfunctional protein?

- A premature stop codon can cause protein production to stop too early. This results in a shortened protein that may be missing important regions needed for its normal function. Because of this, the protein may not fold or function properly.

38. Could a mutation affect protein function without greatly changing protein length?

- **Yes. A mutation can change one amino acid while leaving the overall protein length unchanged.** This happened in our F9 mutation, where c.676C>T changed arginine to tryptophan at position 226, while the predicted protein remained 461 amino acids long.

Even though the length stayed the same, the amino-acid substitution may affect protein function.

39. Could a mutation cause disease without changing the protein sequence? Give a possible molecular mechanism.

- Yes. A mutation can cause disease without changing the protein sequence if it affects gene expression or RNA processing. For example, a mutation in a promoter or regulatory region could reduce the amount of protein produced. A mutation that affects RNA splicing could also change how the RNA is processed even if the coding sequence itself is unchanged.

40. What evidence from your analysis supports the proposed molecular mechanism of your disease?

- Our analysis showed that the F9 c.676C>T mutation changes one nucleotide and changes the codon from CGG to TGG, resulting in an R226W amino-acid substitution. The reading frame and predicted protein length remained unchanged, while the first amino-acid difference occurred at position 226. ClinVar reports that this exact variant has been observed in individuals with hemophilia B and is classified as pathogenic, providing additional evidence that the change in factor IX is related to the disease.

41. Which conclusions are supported directly by your computational results, and which require evidence from published experimental studies?

- The computational results directly support the DNA change, the codon change, the amino acid substitution, the reading frame, and the predicted protein sequence and length. However, our computational analysis alone cannot prove that the altered factor IX protein actually has reduced clotting activity or directly causes the clinical symptoms. Those conclusions require published experimental and clinical evidence. In this case, ClinVar summarizes published evidence linking F9 c.676C>T (p.Arg226Trp) with hemophilia B and reports it as pathogenic.

Part XV. GitHub Commit History

https://github.com/jilduhaylongsod/Hemophilia_B-F9-Mutation-Lab.git

Galaxy Link:

<https://usegalaxy.org/u/jilduhaylongsod/h/duhaylongsod-hemophiliab-f9-mutation-lab>

Part XVI. Limitations

One limitation of the study is that it was mainly conducted through computational analysis using Galaxy, so the effects of the mutations were only predicted based on the DNA and protein sequences. The study did not include laboratory tests to directly determine changes in factor IX structure or clotting activity. It also focused on only one documented mutation and one artificial mutation, so the findings may not represent all possible mutations in the F9 gene. Further laboratory and clinical studies would be needed to confirm the actual effects of these mutations.

Part XVII. Conclusion

The study shows how mutations in the F9 gene can affect the factor IX protein. The documented c.676C>T mutation caused a single amino acid change from arginine (R) to tryptophan (W) at position 226, while the artificial c.71delT mutation caused a frameshift and premature stop codon. The comparison showed that different types of DNA mutations can produce different effects on the protein sequence. Overall, the study showed how changes in the F9 gene can affect factor IX and may contribute to the bleeding problems associated with Hemophilia B.

XV. REFERENCES

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